

CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAM 2008

H2 MATHEMATICS
Paper 2

9740

THURSDAY

28 August 2008

3 hours

Additional material: List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your Name and Class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.

Answer **ALL** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.
The use of a Graphing Calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

At the end of the paper, detach the cover page and attach it to the answer script with the cover page on top.

Name: _____

Class: _____

Question No	Marks	Question No	Marks
1		7	
2		8	
3		9	
4		10	
5		11	
6		Total	

This document consists of 4 printed pages.

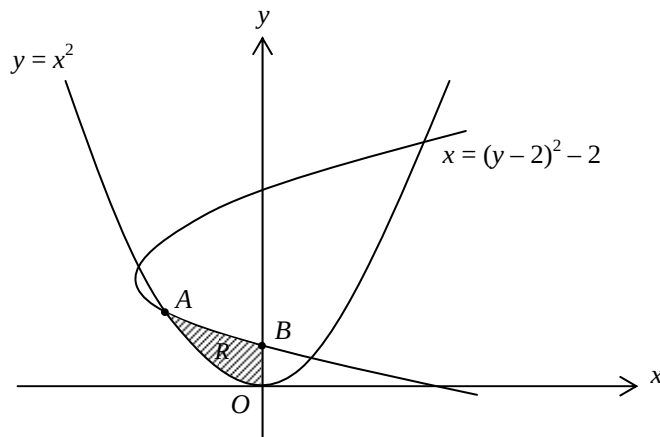
Section A: Pure Mathematics [40 marks]

1. (a) Given that x is small, show that $\sqrt{1 + \sin x} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2$ [2]

(b) Determine the value of a such that $3^x - e^{x \ln a} = 0$. [4]

Given that $e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{3!}$, find the first four terms in the expansion of 3^x as a series in ascending powers of x up to and including x^3 .

(c) A curve $y = f(x)$ passes through the point $(0, 1)$ and satisfies the relation $y^2 \frac{dy}{dx} = \frac{1 + \sin x}{2}$.
By further differentiation of this result, or otherwise, find the Maclaurin's series for y as far as the term in x^3 . [6]
2. Prove by Mathematical Induction, for positive integers $n \geq 1$, that $\sum_{r=1}^n \ln\left(\frac{2r+1}{2r+3}\right) = \ln\left(\frac{3}{2n+3}\right)$. [4]
- Hence express $\sum_{r=6}^{20} \ln\left[\frac{3(2r+1)}{2r+3}\right]$ in the form $A \ln 3 + \ln\left(\frac{B}{C}\right)$, where A, B and C are positive integers to be found. [3]
3. Referred to the origin O , the position vectors of the points A, B and C are $\mathbf{i}$, $\mathbf{i} + \mathbf{j} + \mathbf{k}$ and $6\mathbf{i} - 3\mathbf{j} - 3\mathbf{k}$ respectively. The line l has a vector equation $\mathbf{r} = \lambda(\mathbf{i} + 4\mathbf{j} - 3\mathbf{k})$ where $\lambda \in \mathbb{R}$.
The plane π contains the line l and the point A .
 - (i) Find, in the form $\mathbf{r} \cdot \mathbf{n} = p$, an equation of the plane π . [2]
 - (ii) Find the cosine of the acute angle between lines OA and l .
Hence or otherwise, find the perpendicular distance from A to l . [3]
 - (iii) Find the shortest distance from B to the plane π . [3]
 - (iv) Find the value of m for which the plane π cuts the line BC internally in the ratio $1 : m$ [3]
4. The diagram shows the region R bounded by the two parabolas $y = x^2$ and $x = (y - 2)^2 - 2$ and the y -axis. Find the points indicated A and B in the diagram. [2]
 - (a) Find the area of the region R . [4]
 - (b) Find the volume formed when R is rotated 2π radian about the y -axis. [4]



Section B: Statistics [60 marks]

5. The Ministry of Transport wishes to find out how the recent fuel price hike has affected Singaporean motorists. A list of motorists, in a particular residential estate, is generated and every 50th motorist on the list will be interviewed to get his or her response with regard to this issue.
- (i) Explain why the sample is biased. [1]
 - (ii) What type of sampling method is being used? [1]
 - (iii) If the Ministry is now interested to find out how the fuel price hike has affected Singaporean motorists of different age groups between the ages of 18 and 60, suggest a better sampling method and justify your method. [2]
6. (a) Find the number of different 4-digit numbers that can be formed from the digits 2, 4, 5, 6 if no digit is repeated and the numbers must be divisible by 2. [2]
- (b) Find how many 4-letter code words can be formed from the letters of the word **CUCUMBER**. [4]
7. (a) Ten guests are to sit at a round dinner table. Three of the guests in the list are not comfortable sitting beside either of the other two. If seating is done randomly by a computer, what is the probability that all the three guests are seated separately? [3]
- (b) The probability that John goes to school by MRT, by Bus and by Car are respectively 0.4, 0.5 and 0.1. John is late for school 5% of the time if he takes the MRT. The corresponding figures for Bus and Car are 6% and 1%. Find the probability that John does not take the MRT given that he is late for school. [4]
- During a 5-day week, find the probability that John is late for the second time on the last day. [3]
8. In the production of plastic sheets, small air bubbles occur at random at an average of 1 air bubble in every 2 plastic sheets.
- (i) Show that, in a randomly chosen plastic sheet, the probability that there are at least three bubbles is 0.0144, correct to three significant figures. [2]
 - (ii) Find the most likely number of air bubbles occurring in 5 plastic sheets. [2]
- These plastic sheets are delivered to customers in crates, with each crate containing 15 plastic sheets. A crate is rejected by a customer if it contains at least 2 plastic sheets with at least 3 air bubbles.
- (iii) Find the probability that a randomly chosen crate is rejected. [3]
- 100 randomly chosen crates are delivered to a customer.
- (iv) Find, by using a suitable approximation, the probability that more than 98 crates are not rejected. [3]
9. A student reaches the bus-stop outside his house at 7:20am every morning to take a bus to school and has to reach his school by 7:40am. Assume that the waiting time for his bus is normally distributed with mean 8 minutes and variance 5 minutes², and the journey time is normally distributed with mean 11 minutes and variance 4 minutes².
- (i) Find the probability that he will take more than 20 minutes to reach school (i.e. late for school) on a randomly chosen day. [2]
 - (ii) In a month of 30 days, what is the expected number of days he will be late for school? [1]
 - (iii) Find the latest time he would have to reach the bus-stop outside his house so that the probability of him being late for school is less than 5%. [4]
 - (iv) Find the probability that the average time taken to travel from the bus-stop outside his house (including waiting for the bus) to school in 30 days is between 15 to 20 minutes. [2]

10. Electronic Road Pricing (ERP) is an electronic system of road pricing based on a pay-as-you-use principle. ERP has been effective in maintaining an optimal speed of 60 km per hour for expressways. In general, ERP rates will be increased if the optimal speed is less than 60 km per hour. The relevant authority reviewed the traffic conditions on a particular expressway by measuring the speed of 150 randomly selected cars as they pass a camera. The speed, x km per hour was recorded. The results are summarised by

$$\sum (x - 50) = 1335$$

$$\sum (x - 50)^2 = 23580$$

- (i) Find the unbiased estimates of the population mean and variance. [2]
 - (ii) Test, at the 5% significance level, whether the ERP rates need to be increased for this particular expressway. [4]
 - (iii) Explain, in the context of the question, the meaning of 'at the 5% significance level' stated in (ii). [1]
 - (iv) State, with a reason, whether, in using the above test, it is necessary to assume that the speed of the cars have a normal distribution. [1]
 - (v) Find the smallest level of significance at which the test would result in rejection of the null hypothesis as stated in (ii). [2]
11. Singapore Stock Exchange (SGX) tracked the price of crude oil per barrel from the beginning of 2008. The observations are presented in the table below.

Days in 2008 (t days)	10	31	60	130	155	192	218	245	271
Price of Crude Oil per Barrel (US\$ x)	91	93	92	95	102	120	128	140	175

- (i) Find the equation of the regression line of x on t and calculate the product moment correlation coefficient. [2]
- (ii) On the same diagram, plot a scatter diagram and sketch the regression line of x on t . [2]
- (iii) Using your regression line obtained in (i), calculate the estimated value of x when $t = 130$. Hence comment on the suitability of using a linear model in this case. [2]

Economists from University A propose that a better model is $x = ae^{bt}$, where a and b are positive constants. However economists from University Z claim that their model using $x = c + dt^2$ where c and d are positive constants, is more accurate.

- (iv) Achieve linearity for each of the proposed model by using a suitable transformation and find the corresponding product moment correlation coefficient. [4]
- (v) Hence state with a reason, which University proposed a more appropriate model to describe the situation. [1]

---- THE END ----